



CAPITAL CONSULTANCY

GIS-Based Site Operations ERP

A centralized operating layer for sites, zones, assets, inspections, work orders, employees, attendance, location events, and document control.

CLIENT

Capital Consultancy

PREPARED BY

MaVoid

SCOPE

Phase 1 ERP Build

CONTROLLED FIELD OPERATIONS

Site structure, GIS location, field evidence, approvals, and document control are designed as one connected operating system.

EXECUTIVE PROPOSAL

A controlled operating layer for Capital Consultancy

MaVoid proposes a GIS-enabled ERP platform for site operations, asset records, inspections, work orders, employee attendance, work-event locations, and operational documents.

OPERATING CHAIN

● 01 Site / Zone

● 02 Asset

● 03 Inspection

● 04 Issue / Defect

● 05 Work Order

● 06 Assigned Employee

● 07 Attendance / Location

● 08 Files / Evidence

● 09 Reports

● System controls

Site, zone, building, floor, room, outdoor point, and service-point structure. Asset records hold GPS, QR/barcode, condition, criticality, documents, and history. Inspection and work order records carry evidence, ownership, approvals, labor time, and closure.

● Management gains

One source of truth for site operations and field activity. Clear ownership for every task, defect, approval, and document. Live status across sites, zones, employees, assets, and files. Audit-ready history for inspections, maintenance, attendance, and documents.

SITE STRUCTURE

Locations become the backbone of every record

The system organizes work by physical operating structure, then uses GIS coordinates and map layers to make field activity visible by site, zone, asset, employee event, inspection, and work order.

● Location hierarchy

Site -> Zone / Area -> Building / Facility / Section -> Floor / Level -> Room / Space / Outdoor Point -> Asset. Each level can hold assignments, files, inspections, work orders, and history.

● Map layers

Site boundaries, zones, assets, inspections, issues, work orders, and logged-in staff locations. Filters by site, zone, employee, asset type, inspection status, and work-order status.

● GIS actions

Open asset, inspection, work order, or employee records directly from the map. Capture GPS coordinates from mobile users and validate work events against site rules.

OPERATING SYSTEM MAP

Core modules that run the site operation

The platform is structured around the records operations teams use every day: sites, assets, inspections, work orders, employees, attendance, and documents.

● Dashboard command layer

Management sees total sites, zones, assets, inspection progress, critical issues, open and overdue work orders, attendance exceptions, document expiry, storage usage, and pending approvals.

● Sites and zones

Create sites, zones, regions, buildings, facility areas, floors, rooms, outdoor areas, roads, yards, and service points. Assign assets, employees, inspections, and work orders to exact operating locations. View history across inspections, maintenance, files, employees, and approvals.

● Asset management

Asset ID, asset code, QR/barcode, GPS coordinates, category, type, manufacturer, model, serial number, installation date, warranty, condition, criticality, and status. Linked manuals, certificates, photos, reports, inspection history, work order history, and maintenance history.

● Shared controls

● Asset status

● Operating principle

INSPECTIONS + WORK ORDERS

Failed inspection items become controlled work

The inspection and maintenance modules are one connected flow. Failed checklist items, photos, notes, timestamps, and GPS points create accountable work orders with owners, due dates, evidence, and supervisor closure.

01 Login records time, site, zone, and GPS.

02 Inspection starts and checklist evidence is captured.

03 Failed items create linked defects or work orders.

04 Technician accepts the task and records work evidence.

05 Completion captures photos, files, materials, and labor time.

06 Supervisor verifies closure and history is updated.

● Site inspection module

Templates, checklists, assignments, scheduled dates, start/end times, GPS start/end, photos, notes, pass/fail results, defects, and attached files. Supervisor review covers submitted, approved, rejected, reopened, and completed

● Maintenance and work orders

Manual or inspection-generated work orders linked to sites, zones, locations, assets, and issues. Records include priority, due date, assigned employee/team, GPS points, before/after photos, materials, labor hours.

ATTENDANCE + WORK EVENTS

Operational staff control tied to site work

The Mini HR layer manages employee profiles, roles, teams, site assignments, permissions, login/logout sessions, attendance, GPS validation, and activity history. Location capture is tied to work activity events.

● Employee management

Employee profile, role, team, supervisor, assigned site/zones, contact details, account status, credentials, documents, and attendance history.

● Login and logout tracking

Login time, GPS, site/zone, device, logout time, total duration, attendance status, and exceptions. Flags cover late login, early logout, missing logout, and outside-site login.

● Attendance management

Daily and shift-based attendance by site, team, employee, and date. Statuses cover present, absent, late, early logout, missing logout, outside site, manual adjustment, approved, and rejected.

● Assignments

Assign employees to sites, zones, inspection types, work order types, teams, supervisors, and permissions. Control task access by role, site, zone, and allowed work type.

EVENT-BASED LOCATION CAPTURE

● 01 Login

● 02 Inspection Start

● 03 Inspection End

DOCUMENTS + EVIDENCE

Large operational files stay linked, versioned, and searchable

The file module manages drawings, photos, inspection evidence, maintenance documents, employee files, certificates, permits, warranties, reports, GIS files, CAD drawings, images, videos, Excel sheets, Word documents, and PDFs.

● File linking

Attach files to sites, zones, assets, inspections, work orders, employees, attendance evidence, and reports.

● Large file handling

Chunked uploads, resumable transfer, progress bars, bulk upload, bulk download, object storage compatibility, and file limits.

● Version control

New versions keep previous files, upload date, uploader, version notes, restore history, and approved-version locks.

● Permissions

View, upload, download, edit metadata, approve, reject, delete, restore version, and admin access by role, site, team, and sensitivity.

● Metadata search

File ID, name, type, size, uploader, linked record, site, zone, asset, employee, version, tags, expiry, approval status, and permissions.

● Expiry + approval

Approval workflow, pending/approved/rejected states, renewal reminders, certificate alerts, warranty alerts, and permit alerts.

DELIVERY SCOPE

Phase 1 establishes the operating foundation

Phase 1 delivers the platform foundation and the field workflows required to run sites, assets, inspections, work orders, attendance, map visibility, document control, reporting, and administration.

● Sites, GIS, and assets

Create the location structure, place operational records on the basic GIS map, create asset records, attach files, capture GPS, and view asset history.

● Inspections and work orders

Create templates and checklists, assign inspections, capture GPS, photos, notes, and results, then convert failed items into verified work orders.

● Mini HR and attendance

Employee profiles, teams, roles, site assignments, login/logout, attendance, GPS check-in/check-out, employee map visibility, and attendance reports.

● Files, reports, and admin

Large file upload, attachments, version control, metadata search, permissions, expiry alerts, storage usage, basic reports, rules, templates, and audit logs.

DELIVERY TIMELINE

Estimated implementation plan

The delivery plan is estimated around 12 weeks. Phase durations are shown by workstream, with selected activities running in parallel.

Workstream	Phase duration	Delivery output
Project setup + solution design	2 weeks	Confirm roles, workflows, data model, permissions, environments, and backlog.
Sites, zones + administration	2 weeks	Build location hierarchy, assignments, templates, and admin rules.
GIS, assets + file foundation	3 weeks	Deliver map layers, assets, GPS capture, QR/barcode, metadata, and storage rules.
Inspections + work orders	3 weeks	Build checklists, field evidence, failed-item conversion, assignment, tracking, and closure.
Mini HR + attendance	3 weeks	Deliver employee records, login/logout, attendance, work-event locations, exceptions, and review.
Dashboards, reports + approvals	2 weeks	Connect dashboards, reports, expiry alerts, approval flows, and audit visibility.
QA, UAT + handover	2 weeks	Run acceptance checks, UAT updates, deployment handover, and launch support.
Total estimated duration	Around 12 weeks	Parallel workstreams overlap within the estimated delivery window.

MANAGEMENT VISIBILITY

Reports, governance, and controlled administration

Management reporting turns the operational data model into usable views for assets, inspections, work orders, attendance, employee locations, documents, expiry alerts, and storage usage.

● Main reports

Asset register, asset condition, inspection, failed-inspection, work order, maintenance history, employee attendance, login/logout, employee location, late attendance, missing logout, document register, expired document, and storage usage reports.

● Administration controls

User, role, and permission management. Site and zone setup, asset categories, inspection templates, work order types, attendance rules, shift setup, GPS/geofence rules, document categories, file size limits, notifications, approval workflows, audit logs, and restricted access.

● Controlled build, clear acceptance, auditable handover

Implementation is governed through the module backlog, role rules, approval workflows, acceptance checks, and deployment handover.